MILITARY SPECIFICATION

BAR, PRY, WHEELED TYPE, HAND OPERATED, MA-1, COMMERCIAL

This specification is approved for use within Code 84 Department of the AIR FORCE, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers commercial hand operated wheeled pry bars, having a minimum lifting and moving capacity of 4,000 pounds.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Standards. The following standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

STANDARDS

MILITARY

MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence, For the Selection of
MIL-STD-808	Finishes, Materials, and Processes for Corrosion Prevention and Control in Support Equipment
MIL-STD-889	Dissimilar Metals

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: WR-ALC/MMVRS, ROBINS AFB, GA 31098-5609 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3920

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2.1.2 Other Government documents. The following other Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

COMMERCIAL ITEM DESCRIPTION
MM-L-736 Lumber, Hardwood

(Copies of standards, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The hand operated pry bar shall consist of a lifting plate, wheel assembly (to included wheels, axles and hangers) and a handle. The overall width shall not be greater than 18 inches. The lifting capacity shall not be less than 4,000 pounds.

3.2 Design.

3.2.1 Selection of specifications and standards. Specifications and standards, necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.2.2 Materials. Materials shall be as specified herein. Materials not specifically covered by this specification or referenced document shall be suitable in every respect for this type equipment.

3.2.2.1 Protective treatment. Exposed metal surfaces and materials that are subject to corrosion or deterioration shall be protected in such a manner that shall in no way prevent compliance with the performance requirements of this specification. Protective coatings that will chip, crack or scale with age or extremes of climatic conditions shall not be used.

3.2.2.2 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.2.2.3 Finish and protective coatings. Cleaning, painting, plating, anodic films and chemical treatments shall be in accordance with MIL-STD-808.

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3.2.3 Maintainability. The pry bar shall be designed and constructed to permit removal and replacement of any part or assembly by removing or disconnecting only mounting bolts or fasteners to the part or assembly being removed.

3.2.4 Markings. The following information shall be permanently and legibly stencilled on the top surface of the handle in black characters not less than two inches high and in accordance with MIL-STD-130.

a. "MA-1".
b. Load Capacity "4,000 Pounds".

3.3 Performance.

3.3.1 Load capability. Each pry bar shall be capable of lifting, supporting and moving a minimum of 4,000 pounds. Based on yield strength, the pry bar shall be designed to provide a static safety factor of 1.5.

3.3.2 Lifting plate. The lifting plate shall be constructed of steel, have a width of not less than five inches and a nose height of not less than three inches.

3.3.3 Wheel assembly. The pry bar shall have not less than three wheels of equal size. The wheels shall have a diameter of four inches, plus or minus one inch, and a footprint width of no less than nine inches. The wheels shall be oil resistant rubber or polyurethane covered steel. The axles, wheel hangers and bearings shall be the manufacturer's standard commercial items, provided they meet the requirements specified herein.

3.3.4 Handle. The handle shall be constructed of Number 1 common hardwood of wood species oak, hickory or hard maple in accordance with MM-L-736. The overall length of the handle shall be 72 inches, plus or minus six inches.

3.4 Welding. The surface parts to be welded shall be free from rust, scale, paint, grease and other foreign matter. Welds shall transmit stress without cracking or permanent distortion when the parts connected by the welds are subjected to test, proof and service loadings.

3.5 Screw assemblies. Unless torque is specified, threaded fasteners shall be tight and shall not work loose during testing of service usage.

3.6 Workmanship. The pry bar shall be fabricated and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of welding, brazing, painting, markings and freedom of parts from burrs and sharp edges.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. The contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may use any inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that the item conforms to the prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. First production inspection (see 4.2.1).

b. Quality conformance inspection (see 4.2.2).

4.2.1 First production inspection. The first production inspection shall be applied to the first production item which meets the requirements of this specification. The item shall be identical to the items procured under the terms of the contract. First production inspection shall consist of the examination in 4.3 and all tests in 4.4. Any deficiencies discovered during first production inspection shall be corrected by the contractor at his expense on the first production item and all delivered and undelivered production items. Failure of the first production item to pass the examination and all tests shall be cause for rejection.

4.2.2 Quality conformance inspection. Quality conformance inspection shall be applied to each item being offered for acceptance under the contract. Quality conformance inspection shall consist of the examination in 4.3 and the operational test in 4.4.1.

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4.3 Examination. Each complete pry bar, including the first production item, shall be examined for compliance with the physical and dimensional requirements specified in Section 3 of this specification. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirement or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection. This examination shall be accomplished with checklist showing the results after each requirements.

4.4 Tests. Any malfunction, excessive maintenance, excessive wear, permanent deformation or failure shall be cause for rejection.

4.4.1 Operational test. Pry bars in fulfillment of the contract or order shall be completely assembled, lubricated and otherwise ready for operation. During this operational test, the pry bar shall be demonstrated to lift, support and move a 4,000 pound axle load.

4.4.1.1 Mobility test. The pry bar shall be demonstrated to lift, push and drag a 2,000 pound axle load a distance of 20 feet each direction over level concrete.

4.4.1.2 Performance test. The pry bar shall lift and support a 4,000 pound axle load a series of 10 times.

4.4.2 Load test. The pry bar shall be demonstrated to lift and support a 6,000 pound axle load in accordance with 3.3.1.

4.4.3 Functional test. It shall be demonstrated by functional tests that the pry bar complies with all performances and design requirements of Section 3. These tests may be performed in conjunction with other tests in Section 4.

5. PACKAGING

5.1 PACKAGING. Units shall be packaged for shipment in accordance with the contract.

6. NOTES

6.1 Intended use. The pry bar covered by this specification is intended for use in handling palletized cargo on military aircraft and in the warehouse.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Identification markings required (see 3.2.4).

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6.3 Definitions.

6.3.1 Commercial. A manufacturer's standard product, having been marketed, in production and sold in substantial quantities to the general public for a minimum of one year prior to the date of invitation for bid or request for proposal. An item is "sold in sufficient quantities" only when the quantities regularly sold are sufficient to constitute a real commercial market. Nominal quantities, such as models, samples, prototypes or experimental units, do not meet this requirement. The "general public is a significant number of buyers other than the Government or affiliates of the offeror; the item involved must not be for Government end use. For the purpose of this specification, items acquired for "Government end use" include items acquired for foreign military sales.

6.4 Subject term (key word) listing.

Bar, Pry
Hand Operated
Wheeled Type

Custodian:
Air Force-99

Preparing Activity:
Air Force-84

Project Number:
3920-F177

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-87953 (84)	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* - Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* - Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* - Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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